

* Fundamental equations of logistic Regression:

1) Linear Combination:

$$z = \sum_i w_i x_i + b$$

2) Sigmoid Activation:

$$\hat{y} = \sigma(z) = \frac{1}{1 + e^{-z}}$$

3) Binary Cross-Entropy (BCE)

$$J = -[y \log(\hat{y}) + (1-y) \log(1-\hat{y})]$$

• applying the Chain rule:

$$\frac{\partial J}{\partial w_j} = \frac{\partial J}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial z} \cdot \frac{\partial z}{\partial w_j}$$

• We then compute each derivative:

$$J = -y \log(\hat{y}) - (1-y) \log(1-\hat{y})$$

$$\frac{\partial J}{\partial \hat{y}} = -y \left(\frac{1}{\hat{y}}\right) - (1-y) \cdot \left(\frac{1}{1-\hat{y}}\right) (-1)$$

$$\therefore \frac{\partial J}{\partial \hat{y}} = \frac{-y(1-\hat{y}) + \hat{y}(1-y)}{\hat{y}(1-\hat{y})}$$

$$= \frac{\partial y}{\partial \hat{y}} = \frac{-y + y\hat{y} + \hat{y} - y\hat{y}}{\hat{y}(1-\hat{y})}$$

$$\frac{\partial J}{\partial \hat{y}} = \frac{\hat{y} - y}{\hat{y}(1-\hat{y})}$$

$$\hat{y} = (1 + e^{-z})^{-1}$$

$$\frac{\partial \hat{y}}{\partial z} = -(1 + e^{-z})^{-2} \cdot e^{-z} = \frac{e^{-z}}{(1 + e^{-z})^2}$$

$$\therefore \frac{\partial J}{\partial z} = \left(\frac{1}{1 + e^{-z}} \right) \left(\frac{e^{-z}}{(1 + e^{-z})^2} \right)$$

$$\frac{\partial \hat{y}}{\partial z} = \hat{y} (1 - \hat{y})$$

$$\frac{\partial z}{\partial w_j} = x_j$$

* Combine all :

$$\frac{\partial J}{\partial w_j} = \left(\frac{\hat{y} - y}{\hat{y}(1-\hat{y})} \right) \cdot (\hat{y}(1-\hat{y})) (x_j)$$

$$\therefore \frac{\partial J}{\partial w_j} = (\hat{y} - y) x_j$$

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